

Name: \_\_\_\_\_

**St George Girls High School**

**Trial Higher School Certificate Examination**

**2015**



# **Mathematics**

## **Extension 2**

### **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

### **Total Marks – 100**

#### **Section I – Pages 2 – 5**

##### **10 marks**

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

#### **Section II – Pages 6 – 11**

##### **90 marks**

- Attempt Questions 11 – 16.
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 16.

**Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.**

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**Section I**

**10 marks**

**Attempt Questions 1 – 10**

**Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

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1. Find  $\int \frac{dx}{x^2 - 4x + 13}$

(A)  $\frac{1}{3} \tan^{-1} \left( \frac{x-2}{3} \right) + C$

(B)  $\frac{2}{3} \tan^{-1} \left( \frac{x-2}{3} \right) + C$

(C)  $\frac{1}{3} \tan^{-1} \left( \frac{2x-4}{3} \right) + C$

(D)  $\frac{2}{3} \tan^{-1} \left( \frac{2x-4}{3} \right) + C$

2. The foci of the hyperbola  $\frac{y^2}{8} - \frac{x^2}{12} = 1$  are

(A)  $(\pm 2\sqrt{5}, 0)$

(B)  $(\pm \sqrt{30}, 0)$

(C)  $(0, \pm 2\sqrt{5})$

(D)  $(0, \pm \sqrt{30})$

3. The region bounded by the curves  $y = x^2$  and  $y = x^3$  in the first quadrant is rotated about the  $y$ -axis. The volume of the solid of revolution formed can be found using:

(A)  $V = \pi \int_0^1 \left( y^{\frac{1}{3}} - y^{\frac{1}{2}} \right) dy$

(B)  $V = \pi \int_0^1 \left( y^{\frac{1}{2}} - y^{\frac{1}{3}} \right) dy$

(C)  $V = \pi \int_0^1 \left( y^{\frac{2}{3}} - y \right) dy$

(D)  $V = \pi \int_0^1 (x^4 - x^6) dx$

Section I (cont'd)

4. The five fifth roots of  $1 + \sqrt{3}i$  are:

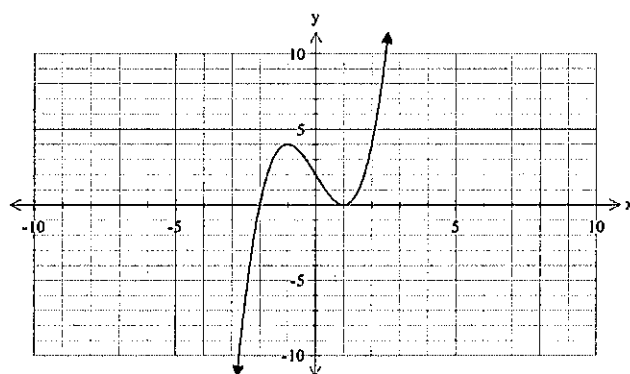
(A)  $2^{\frac{1}{5}} \operatorname{cis} \left( \frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$

(B)  $2^5 \operatorname{cis} \left( \frac{2k\pi}{5} + \frac{\pi}{15} \right), k = 0, 1, 2, 3, 4$

(C)  $2^{\frac{1}{5}} \operatorname{cis} \left( \frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$

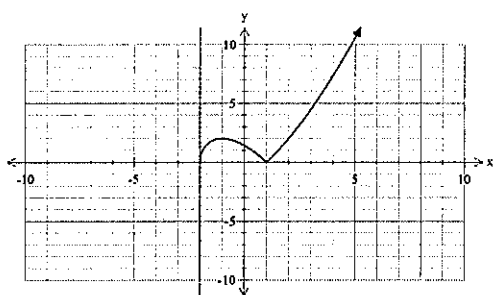
(D)  $2^5 \operatorname{cis} \left( \frac{2k\pi}{5} + \frac{\pi}{30} \right), k = 0, 1, 2, 3, 4$

5. The diagram of  $y = f(x)$  is drawn below.

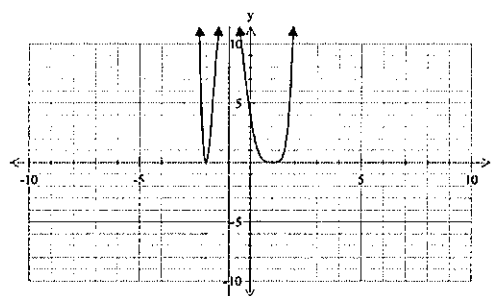


Which of the diagrams below best represents  $y = \sqrt{f(x)}$

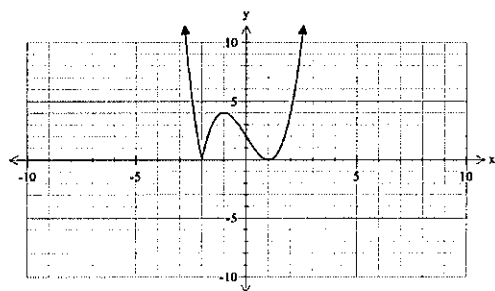
(A)



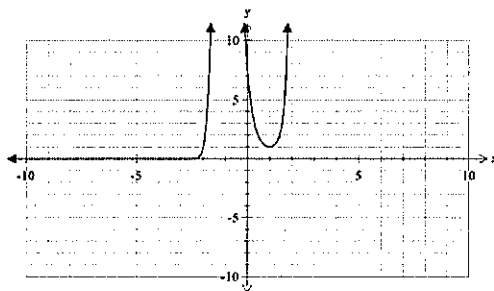
(B)



(C)



(D)

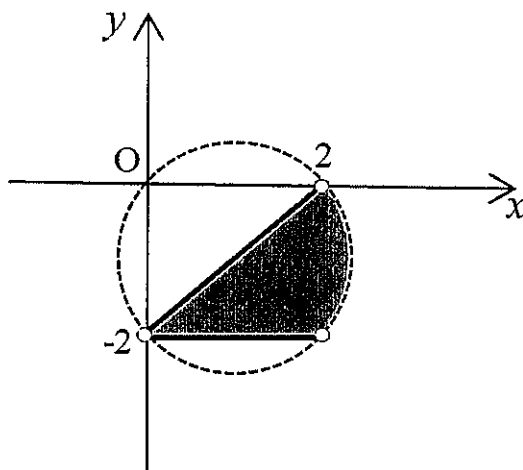


Section I (cont'd)

6. What is the remainder when  $P(x) = x^3 + x^2 - x + 1$  is divided by  $(x - 1 - i)$ ?
- (A)  $-3i - 2$   
(B)  $3i - 2$   
(C)  $3i + 2$   
(D)  $2 - 3i$
7.  $P(x)$  is a polynomial of degree 5 with real coefficients.  $P(x)$  has  $x = -3$  as a root of multiplicity 3 and  $x = i$  as a root. Which of the following expressions is a factorised form of  $P(x)$  over the complex numbers?
- (A)  $P(x) = (x + 3)^3(x - 1)(x + 1)$   
(B)  $P(x) = (x + 3)^3(x - 1)^2$   
(C)  $P(x) = (x + 3)^3(x - i)(x + i)$   
(D)  $P(x) = (x + 3)^3(x - i)^2$
8. Let the point  $A$  represent the complex number  $z$  on an Argand diagram. Which of the following describes the locus of  $A$  specified by  $|z + 3| = |z|$ ?
- (A) Perpendicular bisector of the interval joining  $(0,0)$  and  $(3,0)$   
(B) Perpendicular bisector of the interval joining  $(0,0)$  and  $(-3,0)$   
(C) Circle with a centre  $(0,0)$  and radius of 1.5 units  
(D) Circle with a centre  $(0,0)$  and radius of 3 units
9. A particle of mass  $m$  is moving in a straight line under the action of a force.
- $$F = \frac{m(5 - 7x)}{x^3}$$
- Which of the following equations is the representation of its velocity, if the particle starts from rest at  $x = 1$ ?
- (A)  $v = \pm \frac{3}{x} \sqrt{x^2 - 7x + 5}$   
(B)  $v = \pm \frac{1}{x} \sqrt{-9x^2 + 14x - 5}$   
(C)  $v = \pm 3x \sqrt{x^2 - 7x + 5}$   
(D)  $v = \pm x \sqrt{9x^2 + 14x - 5}$

Section I (cont'd)

10. A region on the Argand Diagram is part of a circle with centre  $(1, -1)$ , as shown below.



Which inequality could define the shaded area?

- (A)  $|z - 1 + i| \leq 1$  and  $0 < \arg(z + 2i) < \frac{\pi}{4}$   
(B)  $|z - 1 - i| < \sqrt{2}$  and  $0 \leq \arg(z - 2i) \leq \frac{\pi}{4}$   
(C)  $|z - 1 + i| \leq 1$  and  $0 < \arg(z + 2i) \leq \frac{\pi}{4}$   
(D)  $|z - 1 + i| < \sqrt{2}$  and  $0 \leq \arg(z + 2i) \leq \frac{\pi}{4}$

End of Section I

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## Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet

**Marks**

a) Let  $A = 3 + 3\sqrt{3}i$  and  $B = -5 - 12i$ . Express each of the following in the form  $x + iy$ :

(i)  $\bar{B}$  1

(ii)  $\frac{A}{B}$  2

(iii)  $\sqrt{B}$  2

b) i) Find the modulus and argument of  $A$ , where  $A = 3 + 3\sqrt{3}i$  2

ii) Hence find  $A^4$  in the form of  $x + iy$ . 1

c) The roots of the polynomial equation  $2x^3 - 3x^2 + 4x - 5 = 0$  are  $\alpha, \beta$  and  $\gamma$ . Find the polynomial equation which has roots:

(i)  $\frac{1}{\alpha}, \frac{1}{\beta}$  and  $\frac{1}{\gamma}$ . 2

(ii)  $2\alpha, 2\beta$  and  $2\gamma$ . 2

d) Find  $\int \frac{dx}{\sqrt{9 + 16x - 4x^2}}$ . 3

**Question 12** (15 marks) Use a SEPARATE writing booklet.

**Marks**

a) Evaluate  $\int_0^{\sqrt{\pi}} 3x \sin(x^2) dx$ . 3

b) (i) Find the values of  $A, B$ , and  $C$  such that: 4

$$\frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{A}{x} + \frac{B}{x - 1} + \frac{C}{x + 2}$$

(ii) Hence find  $\int \frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} dx$

c) Solve the equation  $x^4 - 7x^3 + 17x^2 - x - 26 = 0$ , given that  $x = (3 - 2i)$  is a root of the equation. 3

d) (i) Find the equation of the tangent at the point  $P\left(ct, \frac{c}{t}\right)$  on the rectangular hyperbola  $xy = c^2$ . 2

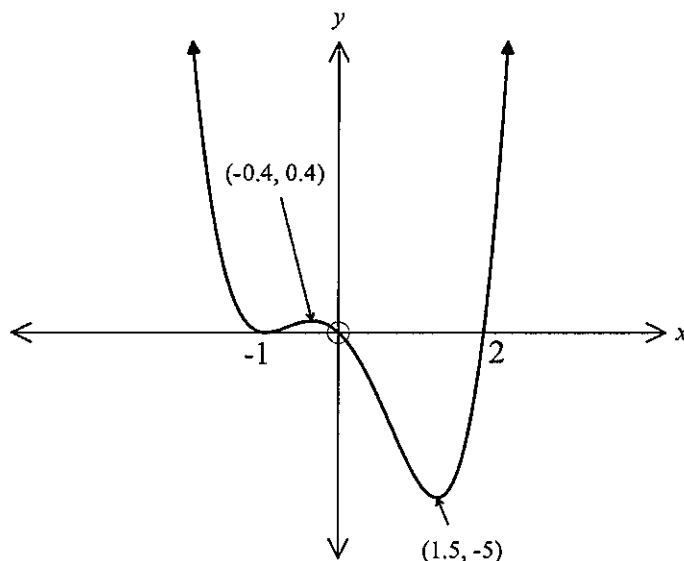
(ii) Find the coordinates of  $A$  and  $B$  where this tangent cuts the  $x$  and  $y$  axis respectively. 2

(iii) Prove that the area of the triangle  $OAB$  is a constant. (Where  $O$  is the origin). 1

**Question 13** (15 marks) Use a SEPARATE writing booklet.

**Marks**

a) The graph of  $y = f(x)$  is shown below.



Draw separate sketches for each of the following:

- |                           |   |
|---------------------------|---|
| (i) $y =  f(x) $          | 1 |
| (ii) $y = \frac{1}{f(x)}$ | 2 |
| (iii) $y^2 = f(x)$        | 2 |
| (iv) $y = e^{f(x)}$       | 2 |
- b) At the start of the observation yesterday, the upper deck of a ship, anchored at Sydney Wharf was 1.2 metres above the wharf at 6:13am, when the tide was at its lowest level. At 12:03pm, at the following high tide, the last observation record shows that the upper deck was 2.6 metres above the wharf. Considering that the tide moves in simple harmonic motion, find:
- |                                                                                                       |   |
|-------------------------------------------------------------------------------------------------------|---|
| (i) At what time, during the observation period, was the upper deck exactly 2 metres above the wharf? | 2 |
| (ii) What was the maximum rate at which the tide increased during this period of observation?         | 2 |
- c) Use the method of cylindrical shells to find the volume of the solid generated by revolving the region enclosed by  $y = 3x^2 - x^3$  and the x axis around the y-axis.

**Question 14** (15 marks) Use a SEPARATE writing booklet

**Marks**

- a) A particle of mass  $m$  kg is dropped from rest in a medium where the resistance to the motion has magnitude  $\frac{1}{40}mv^2$  when the speed of the particle is  $v$  ms<sup>-1</sup>. After  $t$  seconds the particle has fallen  $x$  metres. The acceleration due to gravity is 10 ms<sup>-2</sup>.

(i) Explain why  $\ddot{x} = \frac{1}{40}(400 - v^2)$ . 1

(ii) Find an expression for  $t$  in terms of  $v$ . 2

(iii) Show that  $v = 20 \left(1 - \frac{2}{1+e^t}\right)$ . 1

(iv) Show that  $x = 20 \left[t + 2\ln\left(\frac{1+e^{-t}}{2}\right)\right]$  2

- b) Consider the hyperbola with equation  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ .

(i) Show that the equation of the tangent at the point  $P(a \sec \theta, b \tan \theta)$  has the equation  $bx \sec \theta - ay \tan \theta = ab$ . 2

(ii) Find the equation of the normal at  $P$ . 2

(iii) Find the coordinates of the points  $A$  and  $B$  where the tangent and normal respectively cut the  $y$ -axis. 2

(iv) Show that  $AB$  is the diameter of the circle that passes through the foci of the hyperbola. 3

**Question 15** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- a) Derive the reduction formula:

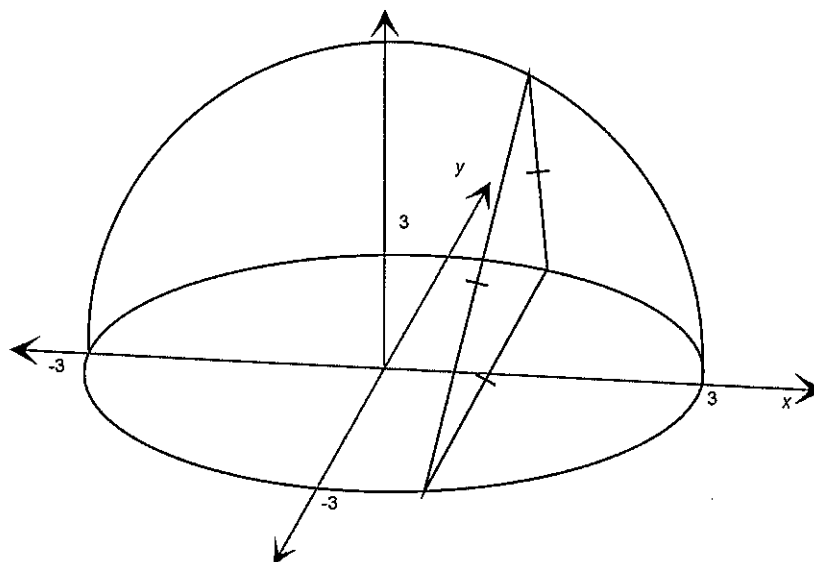
4

$$\int x^n e^{-x^2} dx = -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx$$

and use this reduction formula to evaluate  $\int_0^1 x^5 e^{-x^2} dx$

- b)

4



The diagram above shows a solid which has the circle  $x^2 + y^2 = 9$  as its base. All cross-sections perpendicular to the  $x$  axis are equilateral triangles. Calculate the volume of the solid.

- c) Given that  $x^4 - 6x^3 + 9x^2 + 4x - 12 = 0$ , has a double root at  $x = \alpha$ , find the value of  $\alpha$ .

3

- d) If  $z$  represents the complex number  $x + iy$ , Sketch the regions:

(i)  $|\arg z| < \frac{\pi}{4}$

2

(ii)  $\text{Im}(z^2) = 4$

2

**Question 16** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- a) Show that:  $\frac{\cos A - \cos(A+2B)}{2 \sin B} = \sin(A+B)$ . 3
- b) Consider the area enclosed between the graphs of the hyperbola  $xy = 9$  and the line  $x + y = 10$  in the first quadrant. This area is rotated about the  $x$  axis. By taking a cross-section perpendicular to the axis of rotation and sketching an appropriate diagram, find the volume of the generated solid. 4
- c) Consider the function  $f(x) = \sqrt{3 - \sqrt{x}}$
- (i) Find the domain of  $f(x)$ . 1
- (ii) Show that  $f(x)$  is a decreasing function and deduce the range of  $f(x)$  2
- (iii) Show that  $f''(x) = \frac{6-3\sqrt{x}}{16 [\sqrt{3x-x\sqrt{x}}]^3}$  and find the coordinates of any inflection points. 3
- (iv) Sketch the graph of  $y = f(x)$  and show that  $\int_0^9 \sqrt{3 - \sqrt{x}} \, dx = \frac{24\sqrt{3}}{5}$  2

**End of Examination**

Ext 2

Solutions

$$\begin{aligned}
 1. \quad \int \frac{dx}{x^2 - 4x + 13} &= \int \frac{dx}{x^2 - 4x + 4 + 9} \\
 &= \int \frac{dx}{(x-2)^2 + 9} \\
 &= \frac{1}{3} \tan^{-1} \left( \frac{x-2}{3} \right) + C
 \end{aligned}$$

A

$$2. \quad \frac{y^2}{8} - \frac{x^2}{12} = 1$$

$$a = 2\sqrt{2} \quad b = 2\sqrt{3}$$

$$\therefore b^2 = a^2(e^2 - 1)$$

$$(2\sqrt{3})^2 = (2\sqrt{2})^2(e^2 - 1)$$

$$12 = 8(e^2 - 1)$$

$$\frac{12}{8} = e^2 - 1$$

$$e^2 = \frac{20}{8}$$

$$e^2 = \frac{5}{2}$$

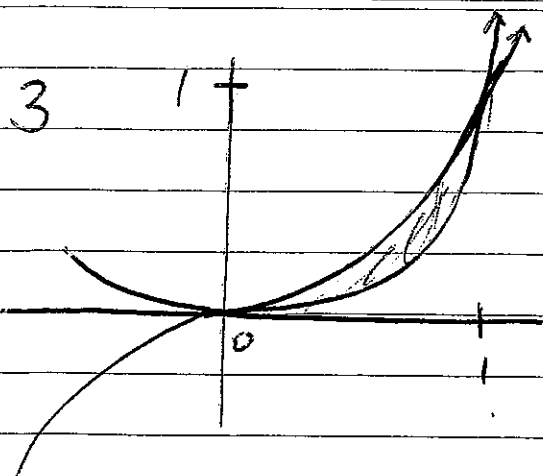
$$e = \frac{\sqrt{10}}{2}$$

$$\begin{aligned}
 \therefore \text{Foci} &= (0, \pm ae) \\
 &= (0, \pm 2\sqrt{2} \cdot \frac{\sqrt{10}}{2})
 \end{aligned}$$

$$= (0, \pm \sqrt{20})$$

$$= (0, \pm 2\sqrt{5})$$

C



$$\text{if } y = x^3, x = y^{\frac{1}{3}}$$

$$y = x^2, x = y^{\frac{1}{2}}$$

$$\therefore V = \pi \int_0^1 \left[ \left( y^{\frac{1}{3}} \right)^2 - \left( y^{\frac{1}{2}} \right)^2 \right] dy$$

$$= \pi \int_0^1 (y^{\frac{2}{3}} - y) dy$$

4. Let  $z = r (\cos \theta + i \sin \theta)$

$$\text{if } z^5 = 1 + \sqrt{3}i$$

$$\text{then } z^5 = r^5 \text{cis } 5\theta$$

$$= 2 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$z = 2^{\frac{1}{5}} \text{cis} \left( \frac{\pi}{15} + \frac{2k\pi}{5} \right)$$

$$\text{for } k=0, 1, 2, 3, 4$$

$$\text{Now } r^5 = \sqrt{1 + (\sqrt{3})^2}$$

$$= \sqrt{1+3}$$

$$= \sqrt{4}$$

$$= 2$$

$$r = 2^{\frac{1}{5}}$$

$$5\theta = \tan^{-1} \sqrt{3}$$

$$= \frac{\pi}{3} + 2k\pi$$

$$\theta = \frac{\pi}{15} + \frac{2k\pi}{5}$$

5. Graph A

6.  $P(x) = x^3 + x^2 - x + 1$

B

Let  $x = 1+i$

$$P(1+i) = (1+i)^3 + (1+i)^2 - (1+i) + 1$$

$$\begin{aligned} &= 2i(1+i) + 2i - 1 - i + 1 \\ &= 2i - 2 + 2i - i \\ &= 3i - 2 \end{aligned}$$

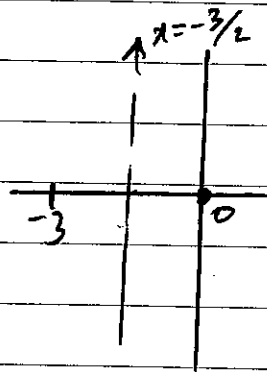
$$\begin{aligned} (1+i)^2 &= 1 + 2i - 1 = 2i \\ (1+i)^3 &= (1+i)^2(1+i) \\ &= 2i(1+i) \end{aligned}$$

7. C

8.  $|z+3| = |z|$

B

$$\begin{aligned} |x+iy+3| &= |x+iy| \\ (x+3)^2 + y^2 &= x^2 + y^2 \\ x^2 + 6x + 9 + y^2 &= x^2 + y^2 \\ 6x + 9 &= 0 \\ 2x + 3 &= 0 \\ x &= -\frac{3}{2} \end{aligned}$$



$\therefore$  Perpendicular bisector  
of the line joining  
(0,0) and (-3,0)

9.  $F = m \left( \frac{5-7x}{x^3} \right)$

~~$m \ddot{x}$~~  =  $m \left( \frac{5-7x}{x^3} \right)$

$\frac{d\left(\frac{1}{2}v^2\right)}{dx} = \frac{5-7x}{x^3}$

$\left. \frac{1}{2}v^2 \right|_0^x = \int_1^x 5x^{-3} - 7x^{-2} dx$

$\frac{1}{2}v^2 = \left[ \frac{5x^{-2}}{-2} + 7x^{-1} \right]_1^x$

$= \left[ -\frac{5}{2x^2} + \frac{7}{x} \right]_1^x$

$= \frac{1}{2} \left[ \frac{14}{x} - \frac{5}{x^2} \right]_1^x$

$= \frac{1}{2} \left[ \left( \frac{14}{x} - \frac{5}{x^2} \right) - (14-5) \right]$

$= \frac{1}{2} \left[ \frac{14}{x} - \frac{5}{x^2} - 9 \right]$

$v^2 = \frac{14x - 5 - 9x^2}{x^2}$

$v = \pm \frac{1}{x} \sqrt{14x - 5 - 9x^2}$

B

10. D.

# Question 11

a) i)  $A = 3 + 3\sqrt{3}i$   $B = -5 - 12i$

$$\bar{B} = \overline{-5 - 12i}$$

$$= -5 + 12i$$

1 mark

(1)

ii)  $\frac{A}{B} = \frac{3 + 3\sqrt{3}i}{-5 - 12i} \times \frac{-5 + 12i}{-5 + 12i}$

$$= \frac{-15 + 36i - 15\sqrt{3}i - 36\sqrt{3}}{25 - 144i^2}$$

1 mark

$$= \frac{-15 - 36\sqrt{3} + i(36 - 15\sqrt{3})}{169}$$

1 mark

(2)

iii)  $\sqrt{B} = \sqrt{-5 - 12i}$

Let  $z = x + iy$  so  $z^2 = -5 - 12i$

Let  $(x + iy)^2 = -5 - 12i$

$$x^2 + 2ixy - y^2 = -5 - 12i$$

$$x^2 - y^2 + 2ixy = -5 - 12i$$

Equate real part

$$x^2 - y^2 = -5 \quad \text{--- (1)}$$

Equate imaginary part

$$2xy = -12 \quad \text{--- (2)}$$

From (2)  $y = \frac{-6}{x}$  sub in (1)

$$x^2 - \left(\frac{-6}{x}\right)^2 = -5$$

1 mark

$$x^4 - 36 = -5x^2$$

$$x^4 + 5x^2 - 36 = 0$$

$$(x^2 + 9)(x^2 - 4) = 0$$

$$x^2 = -9 \text{ or } x^2 = 4$$

$$x = \pm 2 \text{ as } x \text{ is real}$$

Sub this in (2)  $x = 2, y = -3$

$x = -2, y = 3$

$$\therefore \sqrt{-5 - 12i} = 2 - 3i \text{ or } -2 + 3i$$

$$= \pm(2 - 3i)$$

1 mark

(2)

$$\begin{aligned} \text{bi) } \text{mod } r &= \sqrt{(3)^2 + (3\sqrt{3})^2} \\ &= \sqrt{9 + 27} \\ &= \sqrt{36} \\ &= 6 \end{aligned}$$

$$\begin{aligned} \text{arg } B: \tan \theta &= \frac{3\sqrt{3}}{3} = \sqrt{3} \\ \theta &= \frac{\pi}{3} \end{aligned}$$

1

1

(2)

$$\text{ii) } A = 6 \text{ cis } \frac{\pi}{3}$$

$$A^4 = 6^4 \text{ cis } \frac{4\pi}{3}$$

$$= 1296 \text{ cis } -\frac{2\pi}{3}$$

$$= 1296 \left( \cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right)$$

$$= 1296 \left( -\frac{1}{2} + \frac{\sqrt{3}}{2} i \right)$$

$$= -648 \left( 1 - \sqrt{3} i \right)$$

1 mark

(1)

$$\text{c) i) } 2x^3 - 3x^2 + 4x - 5 = 0$$

$$\text{Let } x = \frac{1}{x} \therefore x = \frac{1}{x}$$

$$\therefore \text{ equation is } 2\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 + 4\left(\frac{1}{x}\right) - 5 = 0 \quad 1 \text{ mark}$$

$$\frac{2}{x^3} - \frac{3}{x^2} + \frac{4}{x} - 5 = 0$$

$$2 - 3x + 4x^2 - 5x^3 = 0$$

$$\therefore 5x^3 - 4x^2 + 3x - 2 = 0$$

1 mark

(2)

$$\text{ii) } \text{Let } x = 2x \therefore x = \frac{x}{2}$$

equation is

$$2\left(\frac{x}{2}\right)^3 - 3\left(\frac{x}{2}\right)^2 + 4\left(\frac{x}{2}\right) - 5 = 0$$

1 mark

$$\frac{2x^3}{8} - \frac{3x^2}{4} + \frac{4x}{2} - 5 = 0$$

$$\frac{x^3}{4} - \frac{3x^2}{4} + 2x - 5 = 0$$

$$\therefore x^3 - 3x^2 + 8x - 20 = 0$$

1 mark

(2)

$$\begin{aligned}
 d) \int \frac{dx}{\sqrt{9+16x-4x^2}} \\
 &= \int \frac{dx}{\sqrt{9+4(4x-x^2)}} \\
 &= \int \frac{dx}{\sqrt{9-4(x^2-4x)}} \\
 &= \int \frac{dx}{\sqrt{9-4(x^2-4x+4)+16}} \\
 &= \int \frac{dx}{\sqrt{25-4(x-2)^2}}
 \end{aligned}$$

1 mark for  
completing the  
squares correctly

$$= \int \frac{dx}{\sqrt{4\left(\frac{25}{4} - (x-2)^2\right)}}$$

$$= \frac{1}{2} \int \frac{dx}{\sqrt{\frac{25}{4} - (x-2)^2}}$$

$$\text{Let } u = x-2$$

$$du = dx$$

1 mark

$$= \frac{1}{2} \int \frac{du}{\sqrt{\frac{25}{4} - u^2}}$$

$$= \frac{1}{2} \sin^{-1} \frac{2u}{5} + c$$

$$= \frac{1}{2} \sin^{-1} \frac{2(x-2)}{5} + c$$

1 mark

(3)

## 11 c) Comments

When writing the new equation it is important that it is written as an equation in  $x$ .

1 mark was taken off for an equation not written with respect to  $x$ .

Preferably equations should be written where the highest coefficient is positive.

## 11 d) Care needs to be taken when completing the squares, especially when the quadratic is non-monic.

Many students lost 1 mark.

for not completing the squares correctly.

Q12

$$a) \int_0^{\sqrt{\pi/2}} 3x \sin(x^2) dx$$

$$\text{Let } u = x^2$$

$$du = 2x dx$$

$$\text{when } x=0, u=0$$

$$x = \sqrt{\pi/2}, u = \left(\frac{\sqrt{\pi}}{2}\right)^2 = \frac{\pi}{4}$$

$$= \frac{3}{2} \int_0^{\sqrt{\pi/2}} \sin x^2 \cdot 2x dx$$

$$= \frac{3}{2} \int_0^{\pi/4} \sin u \cdot du$$

1 changing limits + variable

$$= \frac{3}{2} \left[ -\cos u \right]_0^{\pi/4}$$

1 Integral

$$= -\frac{3}{2} \left[ \cos \frac{\pi}{4} - \cos 0 \right]$$

$$= -\frac{3}{2} \left( \frac{1}{\sqrt{2}} - 1 \right)$$

$$= -\frac{3}{2} \left( \frac{1 - \sqrt{2}}{\sqrt{2}} \right)$$

$$= \frac{3\sqrt{2} - 3}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{6 - 3\sqrt{2}}{4}$$

1 Answer

Q126)

$$i) \frac{4x^2 - 3x - 4}{x^3 + x^2 - 2x} = \frac{4x^2 - 3x - 4}{x(x^2 + x - 2)}$$

$$= \frac{4x^2 - 3x - 4}{x(x-1)(x+2)}$$

$$\frac{4x^2 - 3x - 4}{x(x-1)(x+2)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}$$

$$4x^2 - 3x - 4 = A(x-1)(x+2) + Bx(x+2) + Cx(x-1)$$

when  $x=0$

$$-4 = A(-1)(2)$$

$$A = 2$$

$$x=1$$

$$-3 = B(3)$$

$$B = -1$$

when  $x=-2$

$$18 = C(-2)(-3)$$

$$18 = 6C$$

$$C = 3$$

$$ii) \int \frac{4x^2 - 3x - 4}{x(x-1)(x+2)} dx = \int \frac{2}{x} + \frac{-1}{x-1} + \frac{3}{x+2} dx$$

$$= 2 \ln x - \ln |x-1| + 3 \ln |x+2| + C$$

c) As there are real coefficients since  $(3-2i)$  is a factor then  $(3+2i)$  is also a factor

$$\therefore (x - (3-2i))(x - (3+2i)) = x^2 - x(3+2i) - (3x-2i) + (9+4)$$

$$= x^2 - 3x - 2ix + 3x + 2ix + (9+4)$$

$$= x^2 - 6x + 13 \text{ is also a factor}$$

-10-

$$\begin{array}{r}
 x^2 - x - 2 \\
 x^2 - 6x + 13 \overline{) x^4 - 7x^3 + 17x^2 - x - 26} \\
 \underline{x^2 - 6x^3 + 13x^2} \phantom{-x - 26} \\
 -x^3 + 4x^2 - x \phantom{- 26} \\
 \underline{-x^3 + 6x^2 - 13x} \phantom{- 26} \\
 -2x^2 + 12x - 26 \\
 \underline{-2x^2 + 12x - 26} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x^2 - 6x + 13)(x^2 - x - 2) \\
 &= (x^2 - 6x + 13)(x^2 - 2)(x + 1) \\
 &=
 \end{aligned}$$

$\therefore$  Solution to  $x^4 - 7x^3 + 17x^2 - x - 26 = 0$   
is  $3 \pm 2i$ , 2 and -1.

d) i)  $xy = c^2$   
using implicit differentiation

$$y + x \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

when  $x = ct$  and  $y = \frac{c}{t}$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{-\frac{c}{t}}{ct} \\
 &= \frac{-1}{t^2}
 \end{aligned}$$

$$y - \frac{c}{t} = \frac{-1}{t^2}(x - ct)$$

$$t^2 y - ct = -x + ct$$

$$* \quad x + t^2 y - 2ct = 0$$

-11-

(2.11) when  $y=0$ ,  $x+0-2ct=0$

$$x = 2ct$$

$$\therefore A (2ct, 0)$$

when  $x=0$ ,  $0+ty=0$

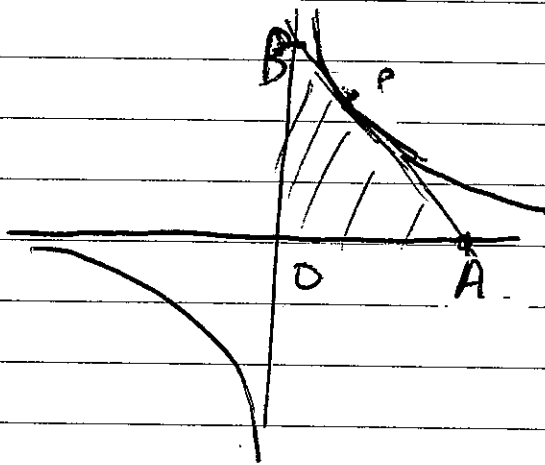
$$y = \frac{2c}{t}$$

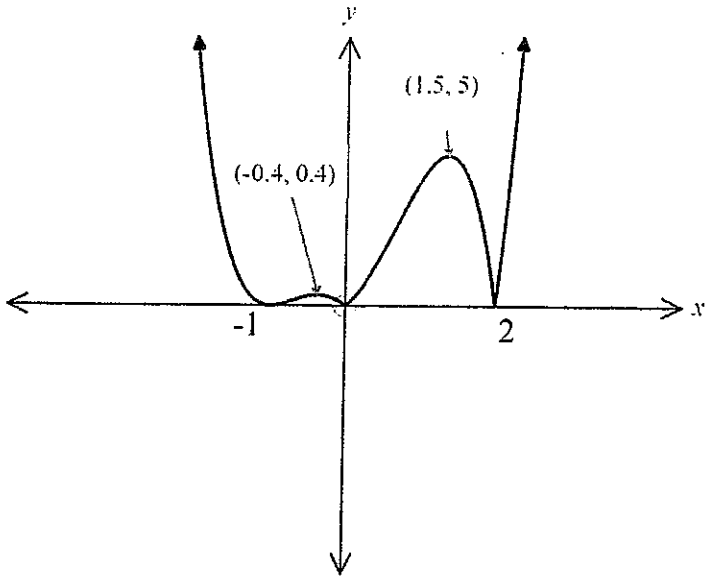
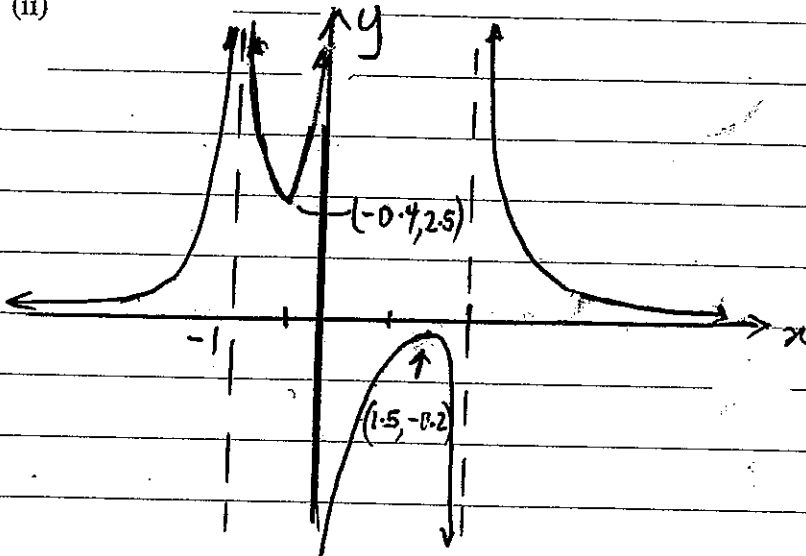
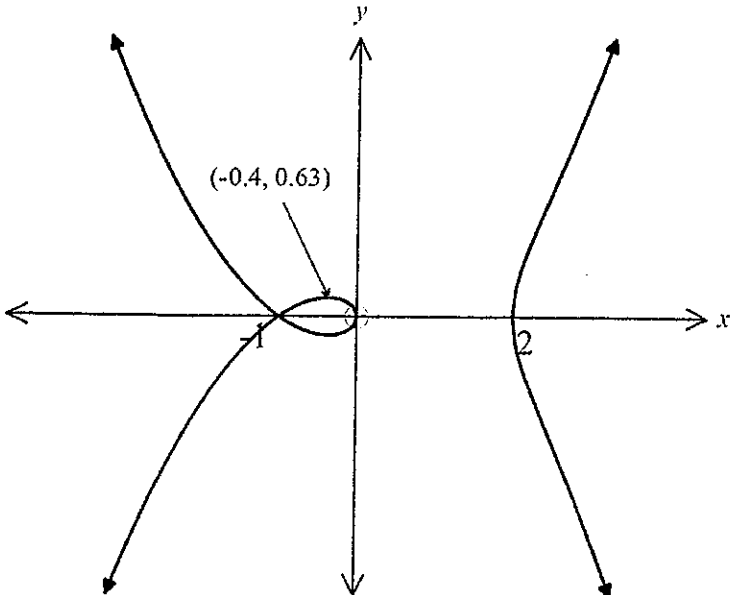
$$\therefore B \text{ is } \left(0, \frac{2c}{t}\right)$$

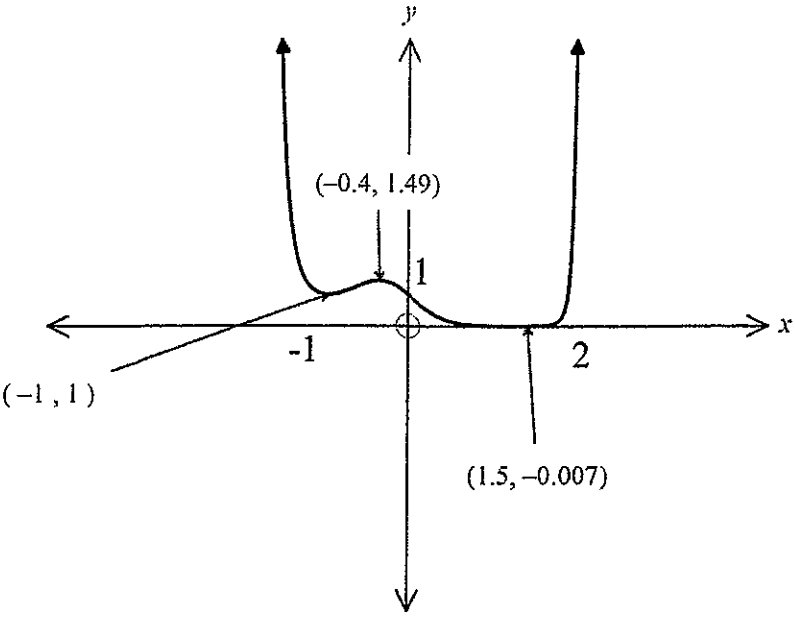
III) Now  $OA = 2ct$

$$OB = \frac{2c}{t}$$

$$\begin{aligned} \text{Area of } \triangle OAB &= \frac{1}{2} \times 2ct \times \frac{2c}{t} \\ &= 2c^2 \end{aligned}$$

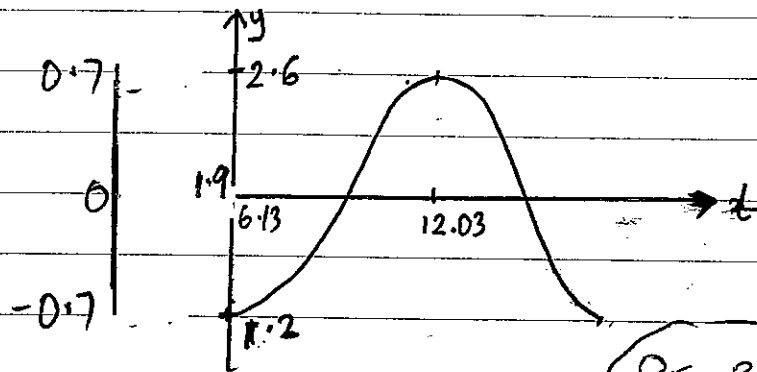


|          |                                                                                     | 2014  |                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solution |                                                                                     | Marks | Allocation of marks                                                                                                                                                                          |
| (i)      |    | 1     | 1 – graph                                                                                                                                                                                    |
| (ii)     |   | 2     | 1 – graph<br>1 – accuracy                                                                                                                                                                    |
| (iii)    |  | 2     | 1 – graph<br>1 – accuracy<br>Care needed to be taken at $(-1, 0)$ and $(0, 0)$<br>At $(-1, 0)$ the graph had to appear as two intersecting lines<br>At $(0, 0) \Rightarrow$ vertical tangent |

| Question 13                                                                                    | 2014                                          |
|------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Solution                                                                                       | Marks Allocation of marks                     |
| <p>(iv)</p>  | <p>2</p> <p>1 - graph</p> <p>1 - accuracy</p> |

Note:  $(0,0)$  was not a discontinuous point, The circle emphasised the origin. Marks were not deducted for this misconception, though.

Q13 b)



$$\text{Period} = 5\text{h}50\text{min} \times 2$$

$$= \frac{35}{6} \times 2\text{h}$$

or

Or period can be found in min

$$\text{period} = 350 \times 2$$

$$= 700 \text{ min}$$

$$T = \frac{35}{3} \text{ h}$$

$$\therefore T = \frac{2\pi}{n}$$

$$= \frac{2\pi}{\frac{35}{3}}$$

$$n = \frac{6\pi}{35}$$

$$\left( \text{or } n = \frac{\pi}{350} \right)$$

$$\text{Centre of motion} = \frac{2.6 + 1.2}{2}$$

$$= 1.9$$

$$\text{Amplitude} = 0.7 \text{ m}$$

As the particle move in SHM  
we use

$$x = 1.9 - 0.7 \cos \frac{6\pi t}{35}$$

1 mark

$$\text{or } x = \left( 1.9 - 0.7 \cos \frac{\pi t}{350} \right)$$

Using  $x = 1.9 - \cos \frac{6\pi t}{35}$

when  $x = 2$

$$2 = 1.9 - 0.7 \cos \frac{6\pi t}{35}$$

$$0.1 = -0.7 \cos \frac{6\pi t}{35}$$

$$-\frac{1}{7} = \cos \frac{6\pi t}{35}$$

$$\frac{6\pi t}{35} = \cos^{-1}\left(-\frac{1}{7}\right)$$

$$t = \frac{35 \cos^{-1}\left(-\frac{1}{7}\right)}{6\pi}$$

$t = 3\text{h } 11\text{min}$  after low tide

Using  $x = 1.9 - 0.7 \cos \frac{\pi t}{350}$

when  $x = 2$

$$2 = 1.9 - 0.7 \cos \frac{\pi t}{350}$$

$$\frac{\pi t}{350} = \cos^{-1}\left(-\frac{1}{7}\right)$$

$$t = \frac{350 \cos^{-1}\left(-\frac{1}{7}\right)}{\pi}$$

$$t = \text{min}$$

$t = 3\text{h } 11\text{min}$

1

$\therefore$  The upper deck was exactly 2m above the wharf at 6.13 am + 3h 11min  
ie 9.24 am.

1 mark

(2)

OR

ii)  $\frac{dx}{dt} = -0.7 \times \frac{6\pi}{35} \sin \frac{6\pi t}{35}$

$\frac{dx}{dt} = -0.7 \times \frac{\pi}{350} \sin \frac{\pi t}{350}$

1

The tide is moving fastest when:-

$$\sin \frac{6\pi t}{35} = 1$$

OR

$$\sin \frac{\pi t}{350} = 1$$

$$\max \frac{dx}{dt} = -0.7 \times \frac{3\pi}{35}$$

$$= \frac{3\pi}{25} \text{ m/h}$$

$$\approx 0.377 \text{ m/h}$$

$$\max \frac{dx}{dt} = -0.7 \times \frac{\pi}{350}$$

$$= \frac{\pi}{500} \text{ m/min}$$

$$\approx 0.00628 \text{ m/min}$$

1

(2)

Q13 b) Alternative solution.

$$x = 0.7 \cos\left(\frac{\pi}{350}t + \alpha\right)$$

To find  $\alpha$  when  $t=0$ ,  $x=-0.7$

$$-0.7 = 0.7 \cos\left(\frac{\pi}{350}(0) + \alpha\right)$$

$$-1 = \cos \alpha$$

$$\alpha = \pi$$

$$\therefore x = 0.7 \cos\left(\frac{\pi}{350}t + \pi\right)$$

when  $x = 0.1$

$$0.1 = 0.7 \cos\left(\frac{\pi}{350}t + \pi\right)$$

$$\frac{1}{7} = \cos\left(\frac{\pi}{350}t + \pi\right)$$

$$\frac{\pi}{350}t + \pi = \pm \cos^{-1}\left(\frac{1}{7}\right) + 2\pi k$$

$$\frac{\pi}{350}t = \pm \cos^{-1}\left(\frac{1}{7}\right) + 2\pi k - \pi$$

$$t = \pm \frac{350}{\pi} \left[ \cos^{-1}\left(\frac{1}{7}\right) + 2\pi k - \pi \right]$$

when  $k=0$

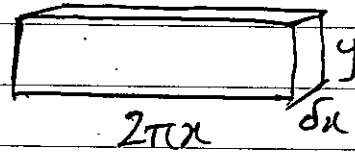
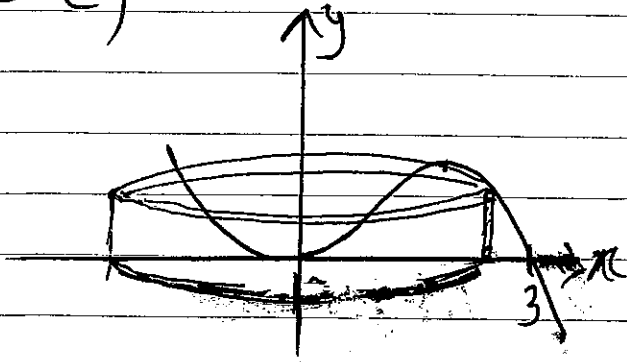
$$t = -190.97 \text{ -- or } t = 190.97 \text{ --}$$

but  $t > 0$

$$\therefore t = 190.97 \text{ min} \div 60 \\ = 3 \text{ h } 11 \text{ min.}$$

(2)

13 c)



$$\begin{aligned} A &= 2\pi xy \\ &= 2\pi x(3x^2 - x^3) \\ \delta V &= 2\pi x(3x^2 - x^3)\delta x \end{aligned}$$

$$V = \sum_{x=0}^3 2\pi x(3x^2 - x^3)\delta x$$

$$= 2\pi \int_0^3 (3x^3 - x^4) dx$$

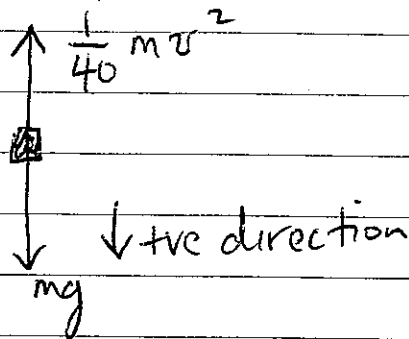
$$= 2\pi \left[ \frac{3x^4}{4} - \frac{x^5}{5} \right]_0^3$$

$$= \frac{243}{10} \pi$$

4

This question was done relatively well,

Q14a)



$$\text{Resultant force} = mg - \frac{1}{40}mv^2$$

$$m\ddot{x} = mg - \frac{1}{40}mv^2$$

$$\ddot{x} = g - \frac{1}{40}v^2$$

$$= \frac{40g - v^2}{40}$$

$$= \frac{400 - v^2}{40}$$

$$= \frac{1}{40}(400 - v^2)$$

ii)  $(v-t)$  rel'p

$$\frac{dv}{dt} = \frac{1}{40}(400 - v^2)$$

$$\frac{dt}{dv} = \frac{40}{400 - v^2}$$

$$= \frac{40}{(20-v)(20+v)}$$

Using partial fractions

$$\frac{40}{(20-v)(20+v)} = \frac{A}{20-v} + \frac{B}{20+v}$$

$$\frac{40}{40} = A(20+v) + B(20-v)$$

when  $v = -20$

$$40 = 40 B$$

$$B = 1$$

when  $v = 20$

$$40 = A(40)$$

$$A = 1$$

$$\therefore \int_0^t dt = \int_0^v \frac{1}{20-v} + \frac{1}{20+v} dv$$

$$t = \left[ -\ln(20-v) + \ln(20+v) \right]_0^v$$

$$= \left[ \ln \left( \frac{20+v}{20-v} \right) \right]_0^v$$

$$= \ln \frac{20+v}{20-v} - \ln \frac{20}{20}$$

$$t = \ln \left( \frac{20+v}{20-v} \right) \quad \text{--- (1)}$$

iii) From (1)

$$e^t = \frac{20+v}{20-v}$$

$$20e^t - ve^t = 20e^t - 20$$

$$v + ve^t = 20e^t - 20$$

$$v(1+e^t) = 20(e^t - 1)$$

$$v = \frac{20(e^t - 1)}{1+e^t}$$

$$= \frac{20e^t - 20}{1+e^t}$$

$$= \frac{20(1+e^t - 1 - 1)}{1+e^t}$$

-20-

$$= 20 \left( \frac{1 + e^t - 2}{1 + e^t} \right)$$

$$v = 20 \left( 1 - \frac{2}{1 + e^t} \right)$$

$$\text{iv) } \frac{dx}{dt} = 20 \left( 1 - \frac{2}{1 + e^t} \right)$$

$$= 20 \left( 1 - \frac{2}{1 + e^t} \times \frac{e^{-t}}{e^{-t}} \right)$$

$$= 20 \left( 1 - \frac{2e^{-t}}{e^{-t} + 1} \right)$$

$$\int_0^x dx = 20 \int_0^t t + 2 \ln |1 + e^{-t}|$$

$$x = 20 \left[ t + 2 \ln |1 + e^{-t}| \right]_0^t$$

$$= 20 \left[ t + 2 \ln (1 + e^{-t}) - 2 \ln 2 \right]$$

$$x = 20 \left[ t + 2 \ln \left[ \frac{1 + e^{-t}}{2} \right] \right]$$

#

14 b) i)  $x = a \sec \theta$       $y = b \tan \theta$

$$\frac{dx}{d\theta} = a \sec \theta \tan \theta$$

and  $\frac{dy}{d\theta} = b \sec^2 \theta$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{d\theta} \times \frac{d\theta}{dx} \\ &= \frac{b \sec^2 \theta \cdot 1}{a \sec \theta \tan \theta} \\ &= \frac{b \sec \theta}{a \tan \theta} \end{aligned}$$

Eqn of tangent

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab (\sec^2 \theta - \tan^2 \theta)$$

$$bx \sec \theta - ay \tan \theta = ab$$

ii)  $m_T = \frac{b \sec \theta}{a \tan \theta}$

$$\therefore m_N = - \frac{a \tan \theta}{b \sec \theta}$$

$$\therefore y - b \tan \theta = - \frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

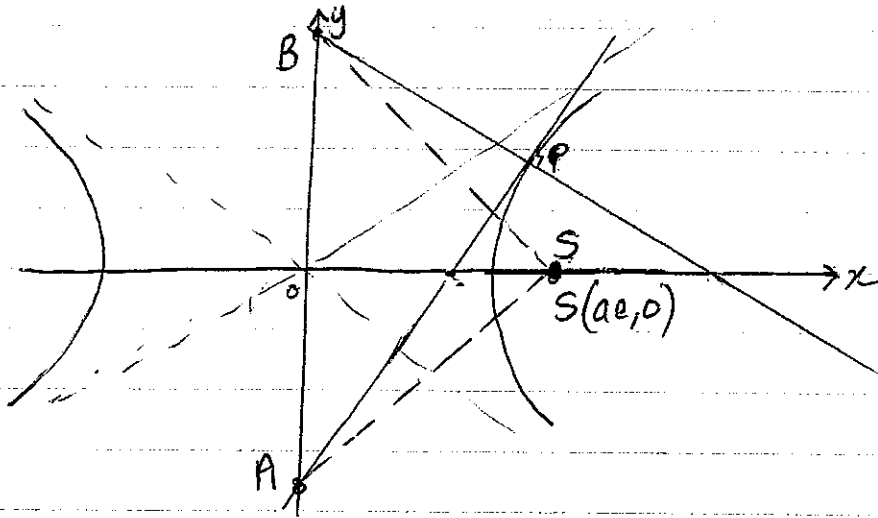
$$by \sec \theta - b^2 \tan \theta \sec \theta = -ax \tan \theta + a^2 \tan \theta \sec \theta$$

$$\div \tan \theta \sec \theta$$

$$\frac{by}{\tan \theta} - b^2 = \frac{-ax}{\sec \theta} + a^2$$

$$\therefore \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

iii)



Tangent cuts the y-axis at A when  $x=0$

$$bx \sec \theta - ay \tan \theta = ab$$

when  $x=0$ ,  $y = \frac{-b}{\tan \theta}$

$$\therefore A \text{ is } (0, \frac{-b}{\tan \theta})$$

$$\text{For } \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

when  $x=0$

$$y = \frac{(a^2 + b^2) \tan \theta}{b}$$

$$\therefore B \text{ is } (0, \frac{(a^2 + b^2) \tan \theta}{b})$$

iv) Focus =  $S(ae, 0)$

If AB is diameter of a circle

RTP,  $\angle ASB = 90^\circ$

Gradient of AS

$$\begin{aligned} m_{AS} &= \frac{0 - \frac{-b}{\tan \theta}}{ae - 0} \\ &= \frac{b}{\tan \theta} \div ae \\ &= \frac{b}{ae \tan \theta} \end{aligned}$$

Gradient of BS

$$\begin{aligned} m_{BS} &= \frac{0 - \frac{(a^2+b^2)\tan \theta}{b}}{ae - 0} \\ &= \frac{-(a^2+b^2)\tan \theta}{b} \div ae \\ &= \frac{-(a^2+b^2)\tan \theta}{abe} \end{aligned}$$

Now

$$\begin{aligned} m_{AS} \times m_{BS} &= \frac{b}{ae \tan \theta} \cdot \frac{-(a^2+b^2)\tan \theta}{abe} \\ &= \frac{-(a^2+b^2)}{a^2e^2} \quad \dots (1) \end{aligned}$$

From  $e^2 - 1 = \frac{b^2}{a^2}$

$$\begin{aligned} a^2e^2 - a^2 &= b^2 \\ a^2e^2 &= a^2 + b^2 \end{aligned}$$

sub in (1)

$$\begin{aligned} m_{AS} \times m_{BS} &= \frac{-(a^2+b^2)}{a^2+b^2} \\ &= -1 \end{aligned}$$

$\therefore \angle ASB = 90^\circ$

$\therefore$  AB is a diameter of a circle passing through S.

# Question 15

a) Let  $I_n = \int x^n e^{-x^2} dx$

$$\int x^n e^{-x^2} dx = \int x^{n-1} x e^{-x^2} dx$$

$$\begin{aligned} u &= x^{n-1} & v' &= x e^{-x^2} \\ u' &= (n-1) x^{n-2} & v &= -\frac{1}{2} e^{-x^2} \end{aligned}$$

$$\begin{aligned} \int x^{n-1} x e^{-x^2} dx &= uv - \int v u' \\ &= x^{n-1} \cdot -\frac{1}{2} e^{-x^2} - \int -\frac{1}{2} e^{-x^2} (n-1) x^{n-2} \\ &= -\frac{1}{2} x^{n-1} e^{-x^2} + \frac{n-1}{2} \int x^{n-2} e^{-x^2} dx \end{aligned}$$

1-mark was used to derive the reduction formula using integration by parts and by only rewriting the integrand as  $\int x^{n-1} x e^{-x^2} dx$  where  $u = x^{n-1}$  &  $v' = x e^{-x^2}$

Notes: No marks were awarded to students who took  $u = x^n$  and  $v' = e^{-x^2}$ . We can't find the integral of  $e^{-x^2}$  to be  $-\frac{1}{2x} e^{-x^2}$ .

## Method 1

Let  $I_n = \int_0^1 x^n e^{-x^2} dx$

$$\begin{aligned} I_5 &= \left[ -\frac{1}{2} x^4 e^{-x^2} \right]_0^1 + 2 \int_0^1 x^3 e^{-x^2} dx \\ &= -\frac{1}{2e} + 2 \left[ \left[ -\frac{1}{2} x^2 e^{-x^2} \right]_0^1 + \frac{3-1}{2} \int_0^1 x e^{-x^2} dx \right] \\ &= -\frac{1}{2e} + 2 \left[ -\frac{1}{2e} + \int_0^1 x e^{-x^2} dx \right] \\ &= -\frac{1}{2e} - \frac{1}{e} + 2 \left[ -\frac{e^{-x^2}}{2} \right]_0^1 \\ &= -\frac{1}{2e} - \frac{1}{e} - \frac{1}{e} + 1 \\ &= 1 - \frac{5}{2e} \end{aligned}$$

1-mark for use of reduction formula

1-mark for subsequent use of reduction formula

1-mark for answer (4)

Method 2

$$\text{Let } I_n = \int_0^1 x^n e^{-x^2} dx$$

$$\begin{aligned} I_5 &= \frac{-1}{2e} + \frac{5-1}{2} I_{5-2} \\ &= \frac{-1}{2e} + 2 I_3 \end{aligned}$$

$$\begin{aligned} I_3 &= \frac{-1}{2e} + \frac{3-1}{2} I_1 \\ &= \frac{-1}{2e} + I_1 \end{aligned}$$

$$\begin{aligned} I_1 &= \int_0^1 x e^{-x^2} dx \\ &= -\frac{1}{2} \int_0^1 2x e^{-x^2} dx \\ &= -\frac{1}{2} \left[ e^{-x^2} \right]_0^1 \\ &= -\frac{1}{2} \left[ \frac{1}{e} - 1 \right] \end{aligned}$$

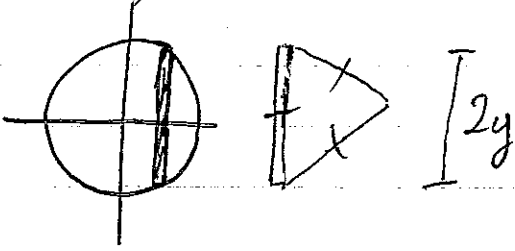
$$\begin{aligned} \therefore I_3 &= \frac{-1}{2e} + \frac{-1}{2} \left( \frac{1}{e} - 1 \right) \\ &= \frac{-1}{2e} - \frac{1}{2e} + \frac{1}{2} \end{aligned}$$

$$\begin{aligned} I_5 &= \frac{-1}{2e} + 2 \left( \frac{-1}{2e} - \frac{1}{2e} + \frac{1}{2} \right) \\ &= \frac{-1}{2e} - \frac{1}{e} - \frac{1}{e} + 1 \\ &= \frac{-5}{2e} + 1 \end{aligned}$$

Method 3

$$\begin{aligned} I_5 &= \int x^5 e^{-x^2} dx \\ &= \left[ -\frac{1}{2} x^4 e^{-x^2} \right]_0^1 - \frac{4}{2} \int_0^1 x^3 e^{-x^2} dx \\ &= \frac{-1}{2e} - 0 - 2 I_3 \\ &= \frac{-1}{2e} - 2 \left[ \frac{-1}{2e} - \frac{2}{2} I_1 \right] \\ &= \frac{-1}{2e} + \frac{1}{e} + 2 \left[ \frac{-1}{2e} + \frac{1}{2} \right] \\ &= \frac{-1}{2e} + \frac{1}{e} - \frac{1}{e} + 1 \\ &= \frac{-5}{2e} + 1 \end{aligned}$$

15 b)



$$\begin{aligned} x^2 + y^2 &= 9 \\ y^2 &= 9 - x^2 \end{aligned} \quad \text{--- (1)}$$

Two methods of finding the area of the cross-section

Method 1

Using  $A = \frac{1}{2} ab \sin C$

$$\begin{aligned} &= \frac{1}{2} \times 2y \times 2y \times \sin 60^\circ \\ &= \frac{1}{2} \times \quad \times \frac{\sqrt{3}}{2} \end{aligned}$$

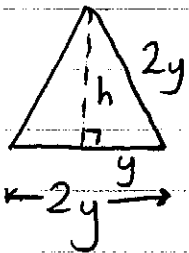
$$A = \sqrt{3} y^2$$

$$A = \sqrt{3} (9 - x^2) \quad \text{from (1)}$$

$$\therefore \delta V = \sqrt{3} (9 - x^2) \delta x$$

Method 2

Using  $A = \frac{1}{2} b h$



$$\begin{aligned} h^2 &= 4y^2 - y^2 \\ &= 3y^2 \end{aligned}$$

$$h = \sqrt{3} y$$

$$\therefore A = \frac{1}{2} \times 2y \times \sqrt{3} y$$

$$= \sqrt{3} y^2$$

$$A = \sqrt{3} (9 - x^2)$$

Now  $\delta V = \sqrt{3} (9 - x^2) \delta x$

$$V \doteq \lim_{\delta x \rightarrow 0} \sum_{x \rightarrow -3}^3 \sqrt{3} (9 - x^2) \delta x$$

$$= \int_{-3}^3 \sqrt{3} (9 - x^2) dx$$

**2** marks for

finding the area of the cross-section

Care needs to be taken when finding the area of the triangle  
Many students took the base to be  $y$  not  $2y$ .

-27-

$$= \sqrt{3} \left[ 9x - \frac{x^3}{3} \right]_{-3}^3$$

1 mark for integral

$$= \sqrt{3} [(27-9) - (-27+9)]$$

$$= \sqrt{3} (18 + 18)$$

$$= 36\sqrt{3} \text{ u}^3$$

1 - answer

4

Q15 c)

$$f(x) = x^4 - 6x^3 + 9x^2 + 4x - 12$$

$$f'(x) = 4x^3 - 18x^2 + 18x + 4$$

Double root occurs when

$$f'(x) = f(x) = 0$$

1 mark  
for using the  
double root thm  
and finding the  
derivative.

Look at factors of 4

(i.e.  $x = \pm 1, x = \pm 2, x = \pm 4$ )

when  $x = 2$

$$\begin{aligned} f'(2) &= 4(2^3) - 18(2^2) + 18(2) + 4 \\ &= 32 - 72 + 36 + 4 \\ &= 0 \end{aligned}$$

$$\begin{aligned} f(2) &= 2^4 - 6(2^3) + 9(2^2) + 4(2) - 12 \\ &= 16 - 48 + 36 + 8 - 12 \\ &= 0 \end{aligned}$$

1 mark for  
testing roots of  
 $f'(x)$

Since  $f'(2) = f(2) = 0$   
then

$(x - 2)$  is a repeated factor

$\therefore x = 2$  is a double root.

1 mark for  
testing in  $f(x)$   
and show  
 $f'(2) = f(2) = 0$   
and stating the  
value of  $x$

Note: Care needs to be taken.

when differentiating and to  
test for a zero we use  
the factors of the constant  
term of  $f'(x)$ .

(3)

# Question 15 d)

-29-

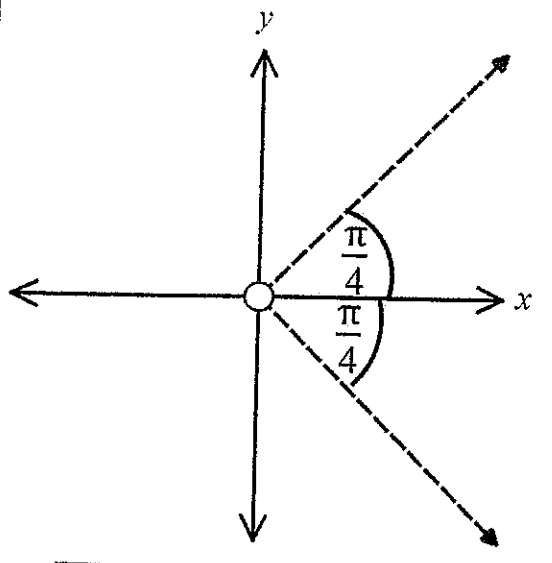
(d) (i)

$$\arg z = \theta$$

$$\text{where } \tan \theta = \frac{y}{x}$$

$$\text{If } |\arg(z)| < \frac{\pi}{4}$$

$$\text{then } -\frac{\pi}{4} < \arg(z) < \frac{\pi}{4}$$



(ii)

$$z = x + iy$$

$$z^2 = (x + iy)^2 = x^2 + 2xyi - y^2$$

$$= x^2 - y^2 + 2xyi$$

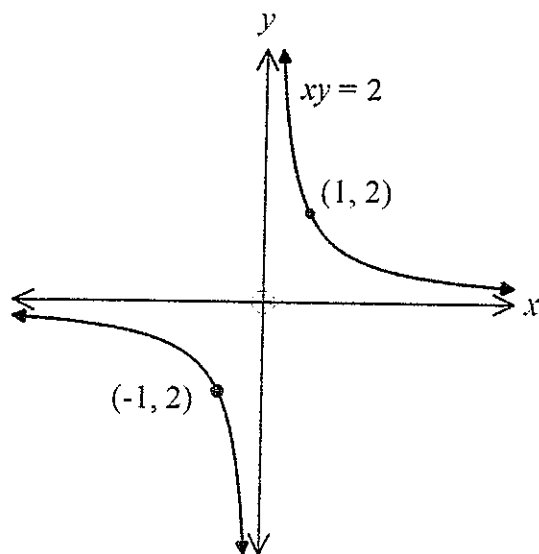
$$\operatorname{Im}(z^2) = 2xy$$

Graph required is  $\operatorname{Im}(z^2) = 4$

$$2xy = 4$$

$$\text{ie } xy = 2$$

$$\text{or } y = \frac{2}{x}$$



1 mark for the graph

1 mark for showing main features.

1 mark

$\frac{1}{2}$  mark (if (0,0) was not removed) 2

1 - determining equation

1 - Graph and points

Note: Generally well done, but always include a valued point on the graph 2

Q16

$$a) \text{ LHS} = \frac{\cos A - (\cos A \cos 2B - \sin A \sin 2B)}{2 \sin B}$$

$$= \cos A - \frac{\cos A \cos 2B}{2 \sin B} + \sin A \sin 2B$$

$$= \cos A - \frac{\cos A (1 - 2 \sin^2 B)}{2 \sin B} + \sin A \cdot 2 \sin B \cos B$$

$$= \cancel{\cos A} - \cancel{\cos A} + \frac{2 \cos A \sin^2 B + 2 \sin A \sin B \cos B}{2 \sin B}$$

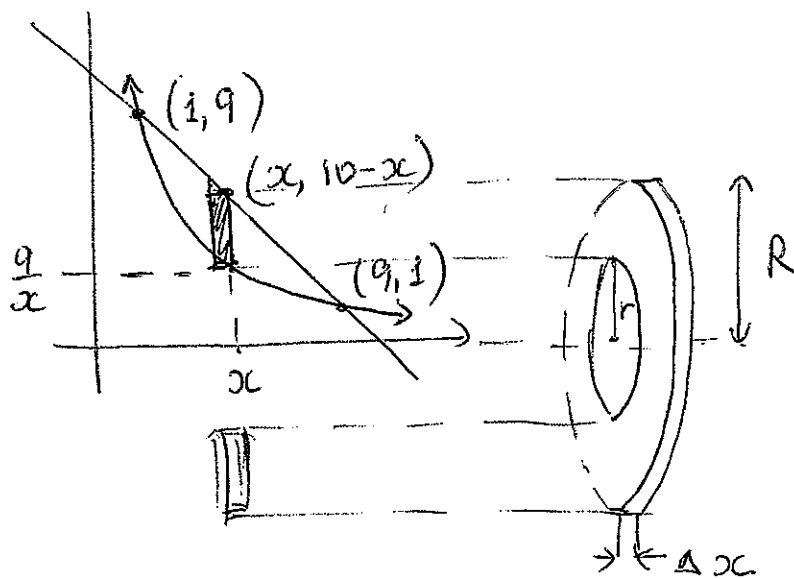
$$= \frac{2 \sin^2 B \cos A + 2 \sin A \sin B \cos B}{2 \sin B}$$

$$= \cancel{2 \sin B} \left( \frac{\sin B \cos A + \sin A \cos B}{\cancel{2 \sin B}} \right)$$

$$= \sin(A + B)$$

$$= \text{RHS}$$

(b) Volume, using the annulus. - 31 -



$$\Delta V \doteq \pi (R^2 - r^2) \Delta x$$

$$R = 10 - x$$

$$r = \frac{9}{x}$$

$$\Delta V \doteq \pi \left( (10 - x)^2 - \left( \frac{9}{x} \right)^2 \right) \Delta x$$

$$V \doteq \pi \sum_{x=1}^9 \left( 100 - 20x + x^2 - \frac{81}{x^2} \right) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \pi \sum_{x=1}^9 \left( 100 - 20x + x^2 - \frac{81}{x^2} \right) \Delta x$$

$$= \pi \int_1^9 100 - 20x + x^2 - 81x^{-2} dx$$

$$= \pi \left[ 100x - \frac{20x^2}{2} + \frac{x^3}{3} - \frac{81x^{-1}}{-1} \right]_1^9$$

$$= \pi \left[ 100x - 10x^2 + \frac{1}{3}x^3 + \frac{81}{x} \right]_1^9$$

$$= \pi \left[ (900 - 810 + 243 + 9) - (100 - 10 + \frac{1}{3} + 81) \right]$$

$$= \pi \left( 342 - 171 - \frac{1}{3} \right) = 170 \frac{2}{3} \pi \text{ u}^3$$

16 Graphs Question

c)  $f(x) = \sqrt{3 - \sqrt{x}} = (3 - x^{\frac{1}{2}})^{\frac{1}{2}}$  2015

(i)  $3 - \sqrt{x} \geq 0$  and  $x \geq 0$

$$3 \geq \sqrt{x}$$

$$9 \geq x \text{ and } x \geq 0$$

$\therefore$  Domain is  $0 \leq x \leq 9$ .

(ii)  $f'(x) = \frac{1}{2} (3 - x^{\frac{1}{2}})^{-\frac{1}{2}} \cdot -\frac{1}{2} x^{-\frac{1}{2}}$  (Chain rule)

$$= -\frac{1}{4} \cdot \frac{1}{\sqrt{x} \sqrt{3 - \sqrt{x}}}$$

$$= -\frac{1}{4} \cdot \frac{1}{\sqrt{3x - x\sqrt{x}}}$$

Since  $\sqrt{3x - x\sqrt{x}} \geq 0$  for all  $x$  in the domain  
 $f'(x) < 0$  for  $0 < x < 9$  and  
 $f'(x)$  is undefined at  $x = 0$  and  $x = 9$ .  
 as  $x \rightarrow 0$  or  $x \rightarrow 9$   $f'(x) \rightarrow -\infty$

$\therefore f(x)$  is a decreasing function

$\therefore f(x)_{\max} = \sqrt{3}$  (when  $x = 0$ )

$f(x)_{\min} = 0$  (when  $x = 9$ )

$$\begin{aligned}
 (11) \quad f''(x) &= -\frac{1}{4} \left( \left( 3x - x^{\frac{3}{2}} \right)^{-\frac{1}{2}} \right)' \\
 &= -\frac{1}{4} \times -\frac{1}{2} \left( 3x - x^{\frac{3}{2}} \right)^{-\frac{3}{2}} \times \left( 3 - \frac{3}{2} x^{\frac{1}{2}} \right) \\
 &= \frac{1}{8} \frac{3 - \frac{3}{2} \sqrt{x}}{\left( \sqrt{3x - x\sqrt{x}} \right)^3} = \frac{1}{16} \frac{6 - 3\sqrt{x}}{\left( \sqrt{3x - x\sqrt{x}} \right)^3}
 \end{aligned}$$

Possible inflexion points:

$$3 - \frac{3}{2} \sqrt{x} = 0$$

$$\frac{3}{2} \sqrt{x} = +3$$

$$\sqrt{x} = \frac{6}{3}$$

$$\sqrt{x} = 2$$

$$x = 4, \text{ as } 0 \leq x \leq 9$$

Check the change of concavity around  $x = 4$ ,

When  $x = 2$

$$f''(x) = \frac{1}{8} \frac{3 - \frac{3}{2} \sqrt{2}}{\left( \sqrt{6 - 2\sqrt{2}} \right)^3} > 0, \text{ as}$$

$$\left( \sqrt{6 - 2\sqrt{2}} \right)^3 > 0$$

$$\text{and } 3 - \frac{3}{2} \sqrt{2} > 0.$$

When  $x = 6$

$$f''(x) = \frac{1}{8} \frac{3 - \frac{3}{2} \sqrt{6}}{\left( \sqrt{6 - 2\sqrt{6}} \right)^3} < 0 \text{ as}$$

$$3 - \frac{3}{2} \sqrt{6} < 0 \text{ and } \left( \sqrt{6 - 2\sqrt{6}} \right)^3 > 0$$

$\therefore$  There is a change in concavity, <sup>34-</sup>  
 When  $x = 4$   $f(x) = \sqrt{3-2} = \sqrt{1} = 1$ .

$\therefore$  The inflexion point is at  $(4, 1)$

$$(iv) A = \int_0^9 \sqrt{3-\sqrt{x}} dx = \int_0^{\sqrt{3}} x dy$$

$$= \int_0^{\sqrt{3}} 9 - 6y^2 + y^4 dy$$

$$= \left[ 9y - \frac{6y^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{3}}$$

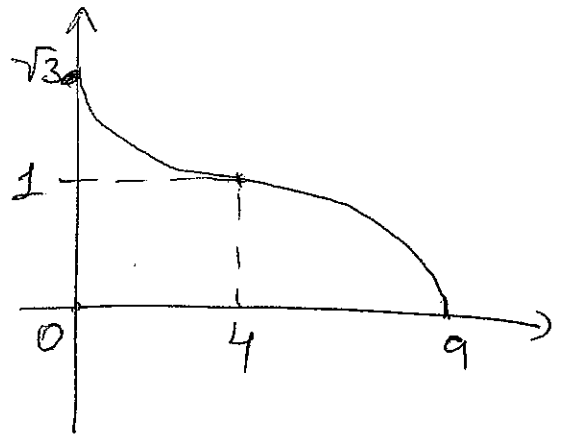
$$= 9\sqrt{3} - 2 \times 3\sqrt{3} + \frac{(\sqrt{3})^5}{5} - 0$$

$$= 9\sqrt{3} - 6\sqrt{3} + \frac{9\sqrt{3}}{5}$$

$$= 3\sqrt{3} + \frac{9\sqrt{3}}{5}$$

$$= \frac{15\sqrt{3} + 9\sqrt{3}}{5}$$

$$= \frac{24\sqrt{3}}{5}$$



$$y = \sqrt{3-\sqrt{x}}$$

$$y^2 = 3-\sqrt{x}$$

$$y^2 - 3 = -\sqrt{x}$$

$$\sqrt{x} = 3 - y^2$$

$$x = (3 - y^2)^2$$

$$x = 9 - 2 \times 3 \times y^2 + y^4$$

$$x = 9 - 6y^2 + y^4$$

# Examiner's Comments.

## SECTION 1 Q1-10

- Generally well done, however Q2, 3 and 9 caused some difficulty.

10% of candidates were incorrect on Q9  
18% " " " " " Q2  
21% " " " " " Q3

## SECTION 2:

### QUESTION 12:

- (a) When making a substitution the integrand should contain one variable only.
- (b) (i) 'Find the values' requires students to set out their working and NOT use some short-cut method to simply write down the values of A, B and C.
- (iii) Well done
- (c) Well done
- (d) ∴ Generally well done.

#### QUESTION 14:

- (a) (i), (ii) and (iii) well done  
(iv) Some students unable to integrate  
simply arrived at the answer magically.
- (b) (i), (ii) and (iii) well done.  
(iv) many students failed to realise that  
the solution involved  $m_1 \times m_2 = -1$

#### QUESTION 16:

- (a) well done
- (b) Generally well done although some  
students wrote  $\left[(10-x) - \frac{x^2}{9}\right]^2$   
rather than  $(10-x)^2 - \left(\frac{x^2}{9}\right)^2$
- (c) (i) Some students failed to realise there  
are TWO parts to this question  
ie  $\sqrt{x} \Rightarrow x \geq 0$  and  $\sqrt{3-\sqrt{x}} \Rightarrow 3-\sqrt{x} \geq 0$
- (ii) Too many students failed to realise  
that  $y \geq 0$
- (iii) Finding  $f''(x)$  and solving  $f''(x) = 0$   
caused few problems. However, too  
many students failed to test for  
point of inflexion.

(iv) Sketches were generally quite poor. Students did not analyse  $f'(u)$  at  $x=0$  and  $x=9$ . Even those who did show that  $f'(u)$  is undefined at  $x=0, 9$  then failed to interpret this correctly in their graphs.

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